Steam power plants

Module-3

Steam power plant

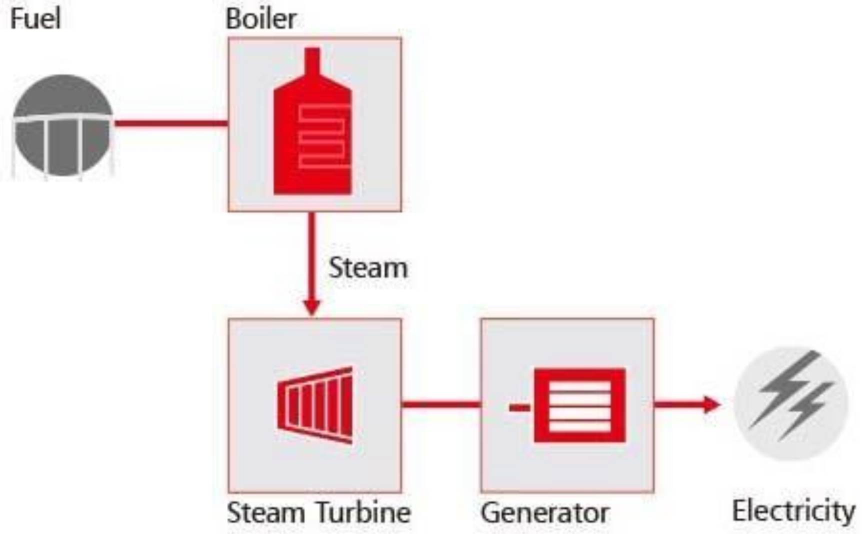
- A steam power plant is **a thermal power station that converts the heat energy of fuel into electricity**.
- The steam power plant is also called a Thermal Power plant.
- The fuel can be coal, oil, gas, or other fossil fuels
- The steam power plant uses the Rankine cycle.

How it works

- The boiler generates steam at high pressure and temperature.
- The steam turbine converts the heat energy of steam into mechanical energy.
- The generator converts the mechanical energy into electric power.

Steam power plant

- A steam power plant consists of a boiler, steam turbine and generator, and other auxiliaries.
- The boiler generates steam at high pressure and high temperature.
- The steam turbine converts the heat energy of steam into mechanical energy.
- The generator then converts the mechanical energy into electric power.



https://power.mhi.com/products/conventional

Classification of steam power plant

- Conventional steam power plants: These plants use fossil fuels, such as coal, oil, or natural gas, to heat water and create steam.
- Nuclear steam power plants: These plants use nuclear fission to heat water and create steam

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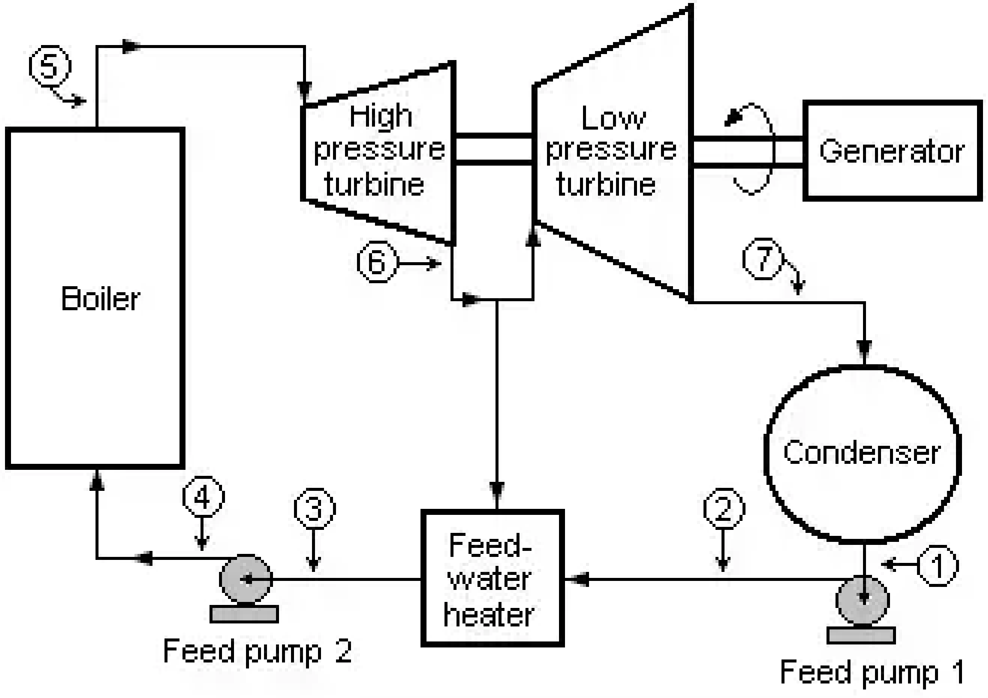
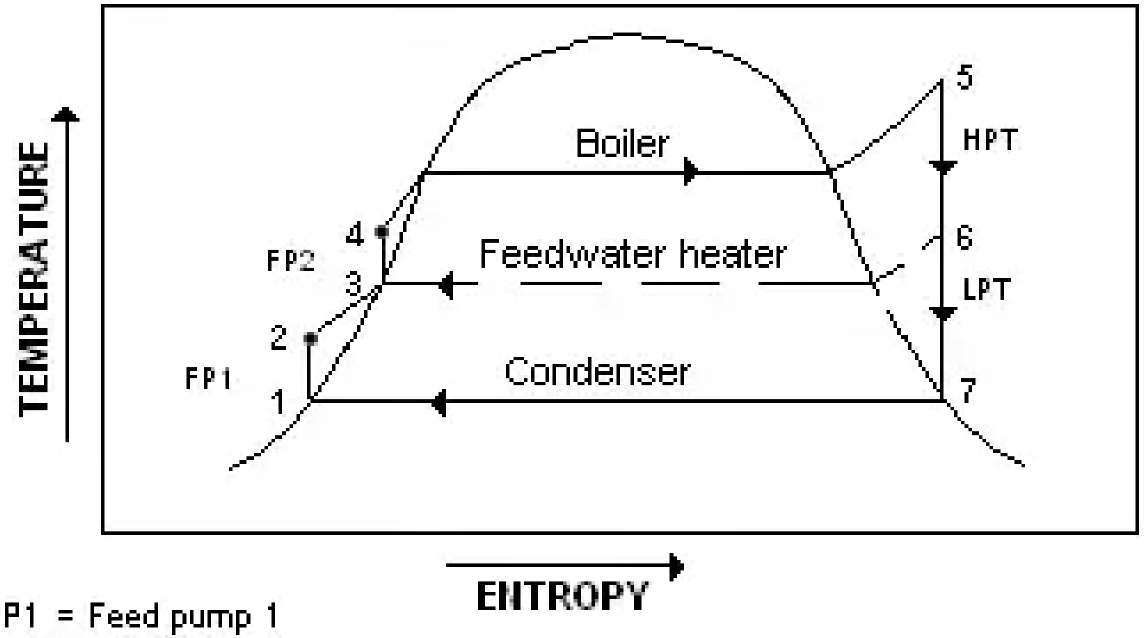
- Central station: Electrical energy from these plants are meant for general sale. These stations are condensing type
- Industrial power stations: Electricity is produced by industries for their own use. Normally these power stations are non- condensing

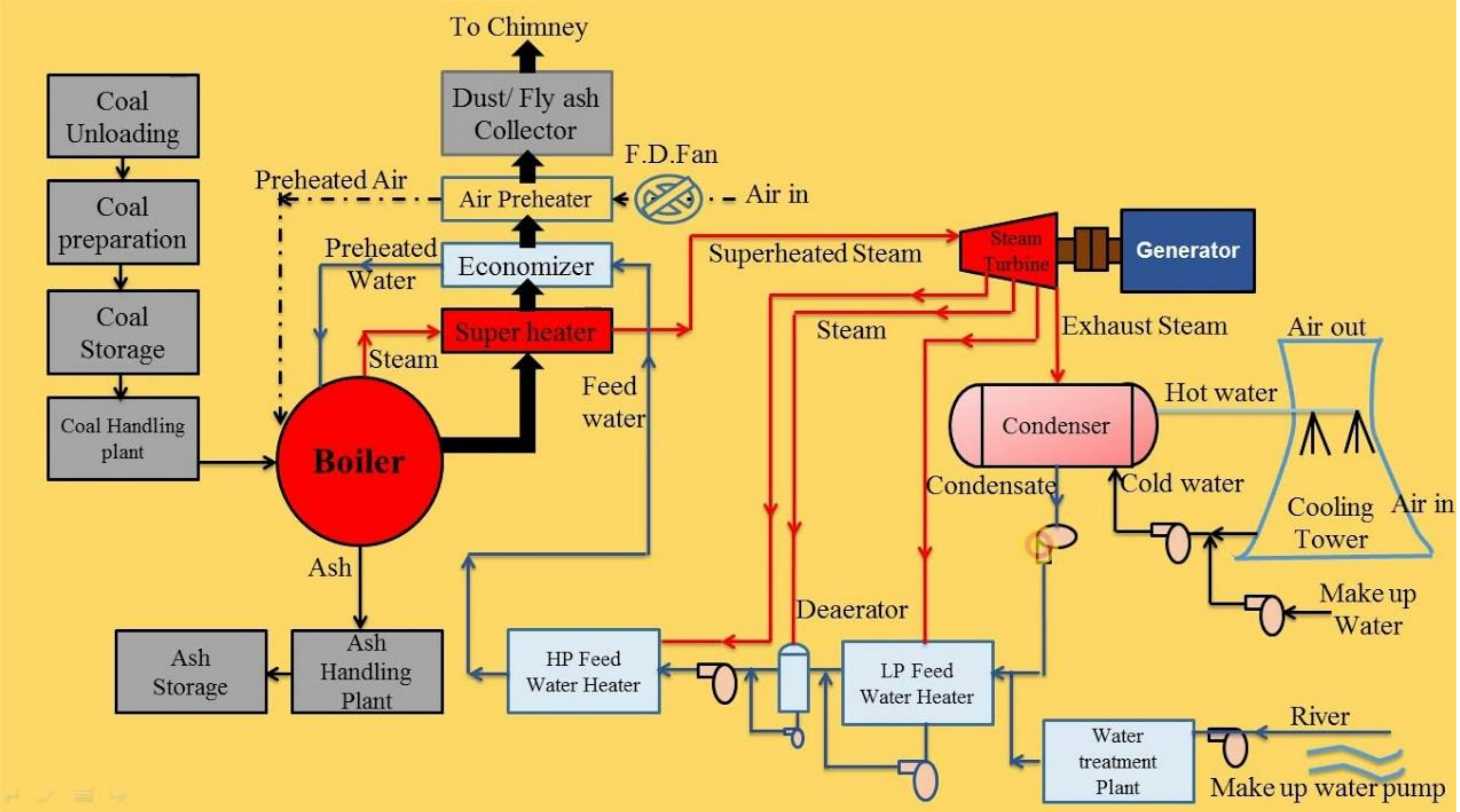
Diagram of Steam Power Plant Running on Rankine Cycle





FP1 = Feed pump 1
FP2 = Feed pump 2
HPT = High pressure turbine
LPT = Low pressure turbine

Layout



To Chimney
Dust/ Fly ash Collector
Coal Unloading
Coal preparation
Coal Storage
Coal Handling plant
Preheated Air
Air Preheater
F.D.Fan
Air in
Preheated Water
Economizer
Super heater
Steam
Boiler
Feed water
Superheated Steam
Steam Turbine
Generator
Steam
Exhaust Steam
Air out
Hot water
Condenser
Condensate
Cold water
Cooling Tower
Air in
Make up Water
Ash
Ash Storage
Ash Handling Plant
HP Feed Water Heater
Deaerator
LP Feed Water Heater
Water treatment Plant
River
Make up water pump

Layout

- **The layout of the steam power plant consists of the following parts.**
- Coal and Ash Handling Unit
- Boiler
- Superheater
- Steam Turbine
- Generator
- Condenser
- Economizer
- Feed Pump
- Cooling Tower
- Chimney

Components of Steam Power Plant

Coal and Ash Handling Unit:

- Before feeding the coal to the furnace, It is to be converted into the pulverized form and after the combustion, the ash is collected in the ash handling unit.

Boiler:

- The equipment used for producing steam is called **Boiler** or Steam Generator.
- The steam generated is used for:
- Heating
- Power generation
- Utilization in industries like sugar mills, chemical industries, etc.

Components of Steam Power Plant

Superheater:

- The steam is taken out from the boiler and is superheated so that, the steam should be free from the water molecules.

Steam Turbine:

- The steam which is free from water molecules are used to strike the blades of the turbine so that large amount of power can be generated.

Generator:

- A generator that converts one form of energy to another is attached to the rotor of the turbine and as the turbine rotates, it also rotates with the speed of the turbine.

Components of Steam Power Plant

Condenser:

- The steam after striking on the blades of the turbine enters into the condenser where it gets condensed with the help of cold water from the cooling tower.

Economizer:

- The economizer adds the feed water temperature.

Feed Pump:

- This feedwater is to be sent to the boiler using a feed pump only.

Components of Steam Power Plant

Cooling Tower:

- The water that is used to condense the steam in the condenser was supplied from the cooling tower.

Chimney:

- It is the last stage where only the air particles will enter into the atmosphere by reducing the heat from steam by the usage of the water sprinklers.

Working Principle of Steam Power Plant

A Steam Power Plant working phenomena will be discussed using 3 circuits:

1. Furnace Circuit
2. Steam Circuit
3. Cooling Water circuit

1. Furnace Circuit:

- Before feeding to the furnace, the coal is to be converted into the pulverized form and after combustion, the ash is to be collected into the Ash Pit (Ash handling unit).

2. Steam Circuit:

- The steam is taken from the boiler and is superheated so that, the steam present should be free from the water molecules. Thus the superheated steam strikes the turbine blades with a high speed which makes the turbine rotate.
- A generator is attached to the rotor of the turbine and as the turbine rotates, it also rotates with the speed of the turbine. The generator converts one form of energy to another i.e. mechanical energy of the turbine into electrical energy.
- After striking the turbine, the steam enters into the condenser where it gets condensed with the help of cold water from the cooling tower.

3. Cooling Water Circuit:

- The water that is used to condense the steam in the condenser was supplied from the cooling tower. The hot steam cannot be released directly into the atmosphere from the chimneys and by doing so, the hot steam will affect the atmosphere.
- For this reason, the hot steam along with water particles was released from the larger heights in the cooling tower, so that the water will fall down and the air particles will escape into the atmosphere.

Advantages of Steam Power Plant

- The coal for the combustion is to be transported via railways and roadways.
- The fuel used is quite cheaper compared to other modes of power generation processes.
- The setup cost is also very low compared to others.
- Space accommodation is less compared to the hydropower plant.

Disadvantages of Steam Power Plant

- The steam power plant produces smoke and fumes at the end of the combustion process which can pollute the environment.
- Time Consumption is more for those power plants which are away from the Coalfield.

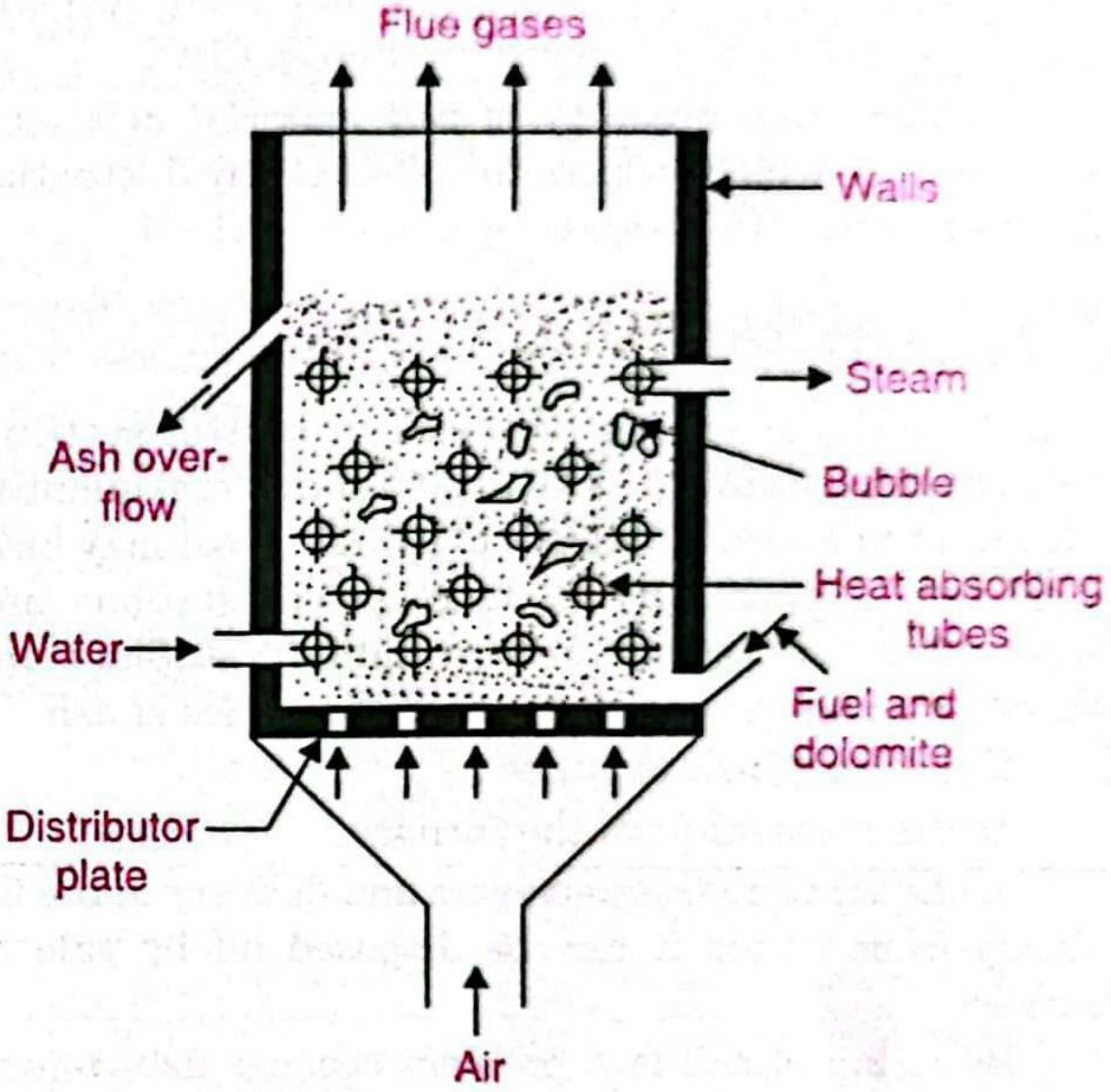
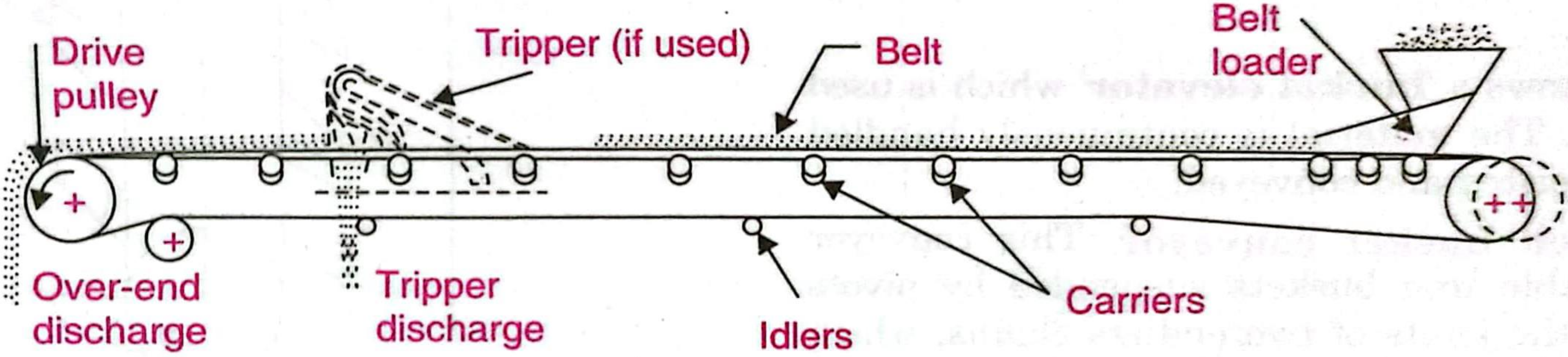
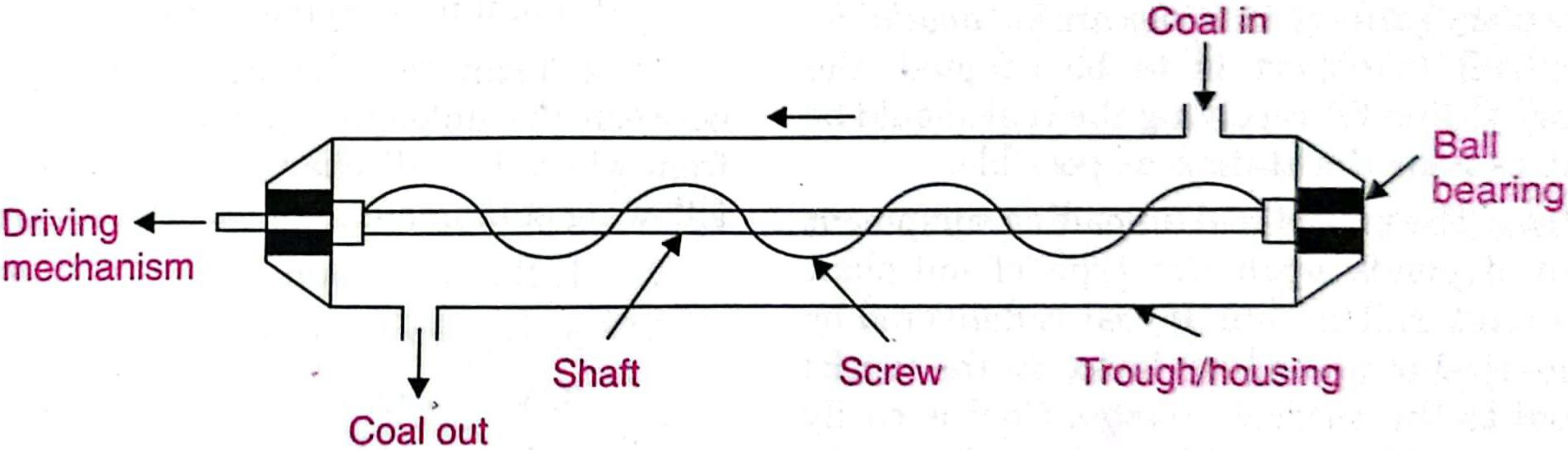


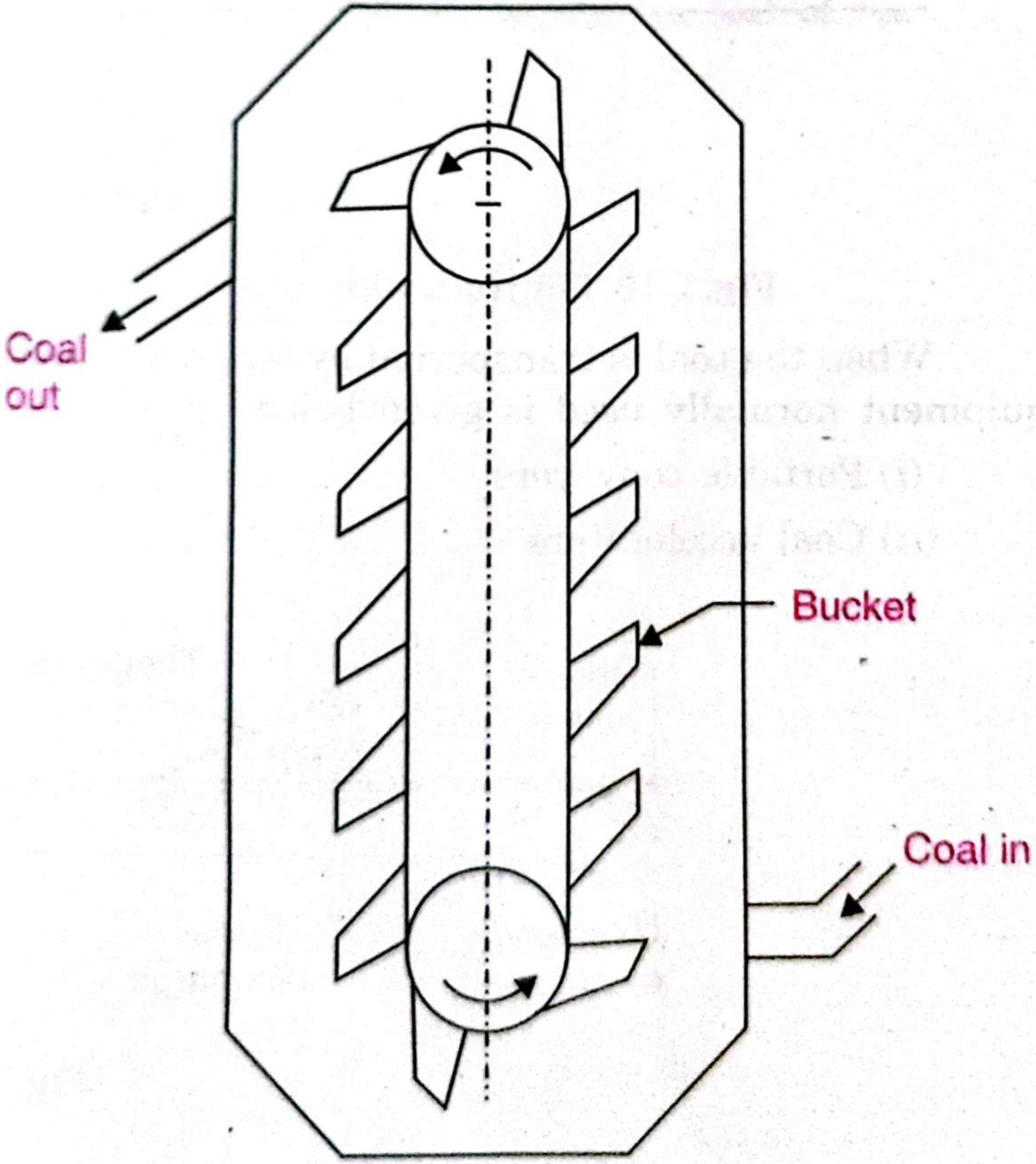
Fig. 3.37. Basic FBC system.



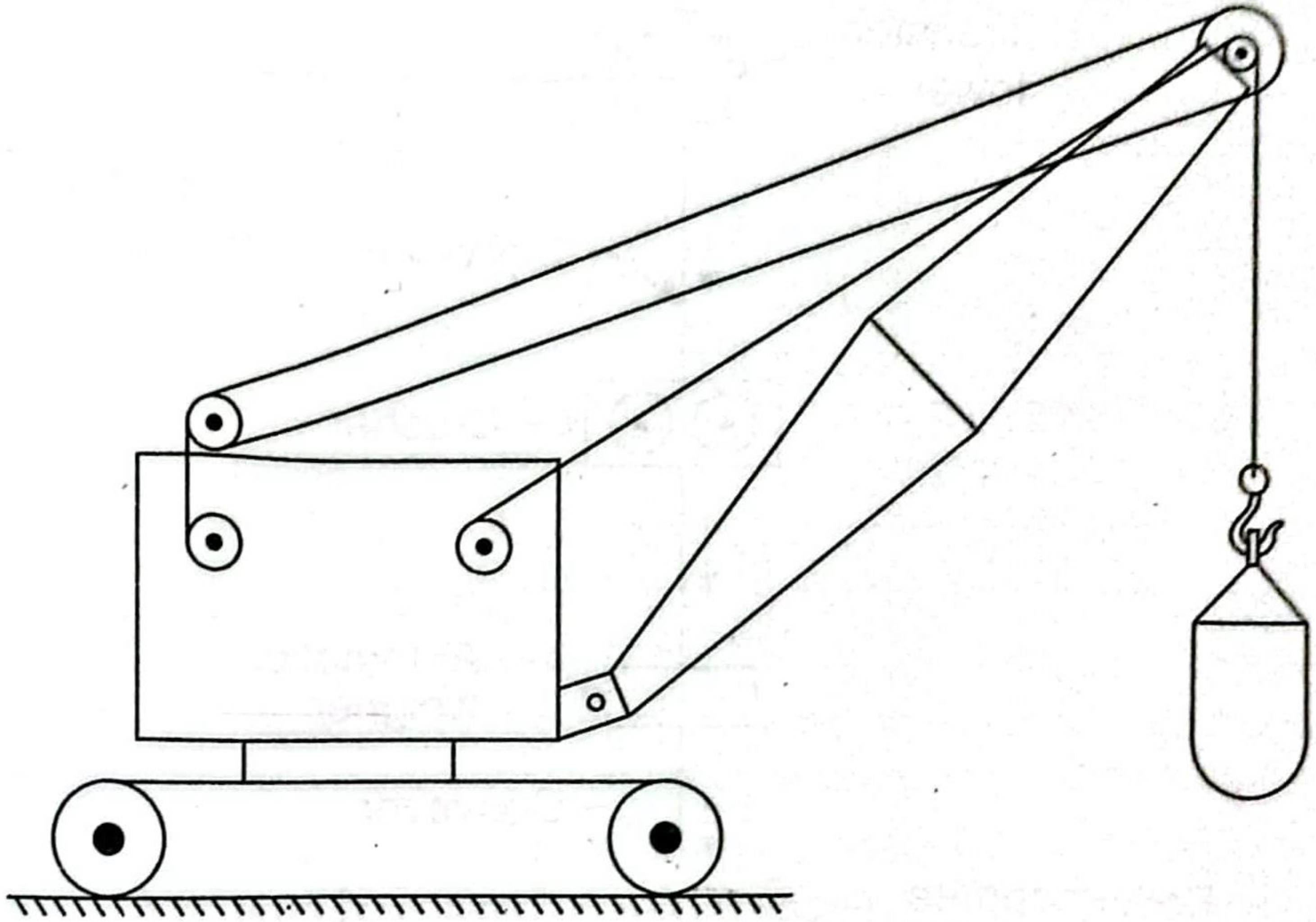
Belt conveyor.



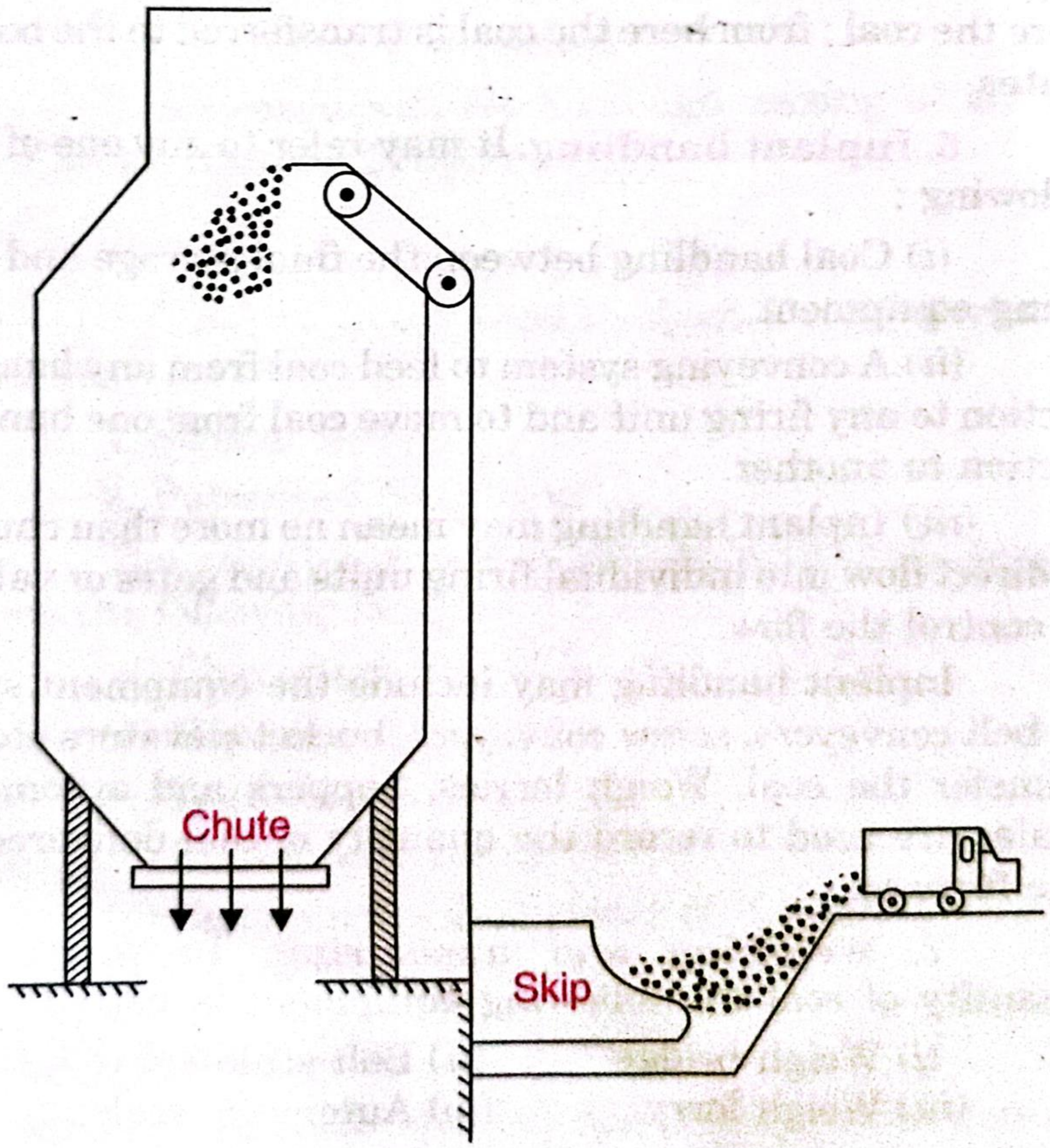
Screw conveyor.



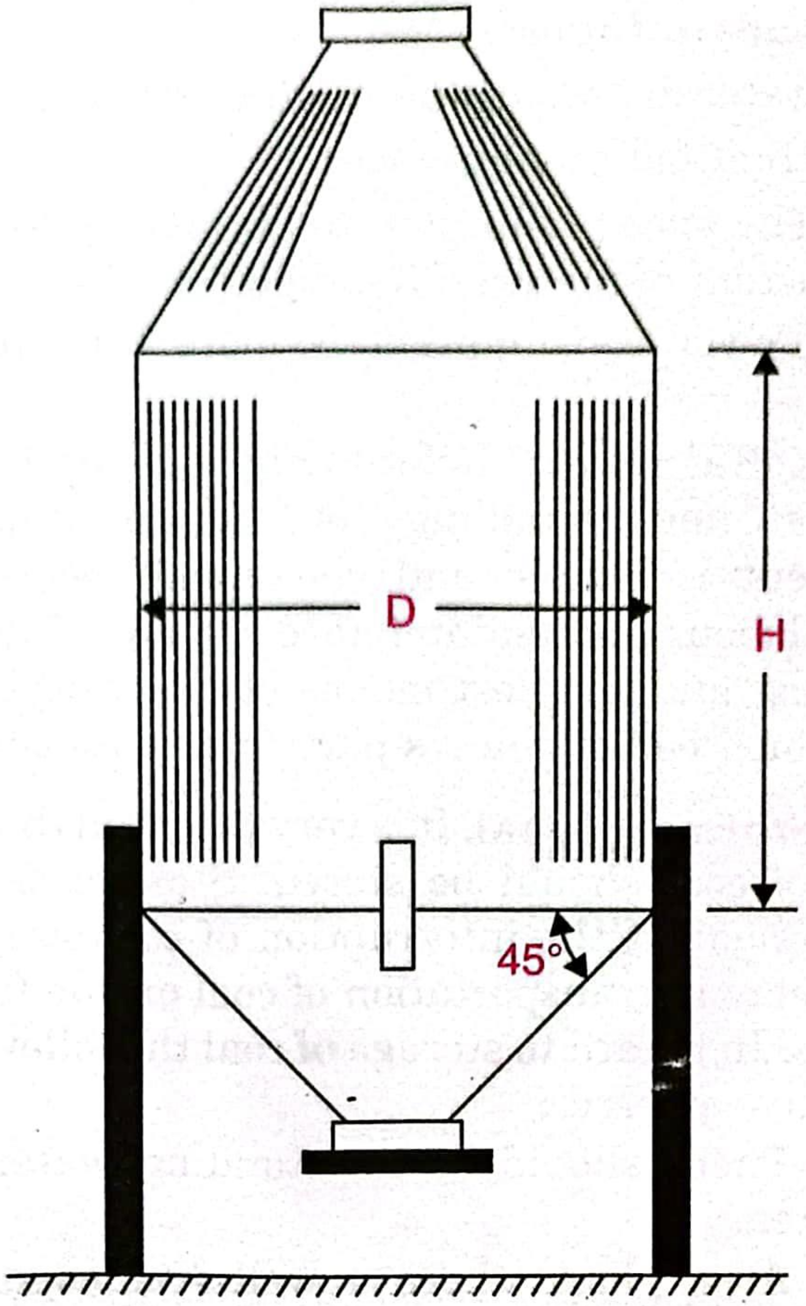
Bucket elevator.



Grab bucket conveyor.



Skip hoist.

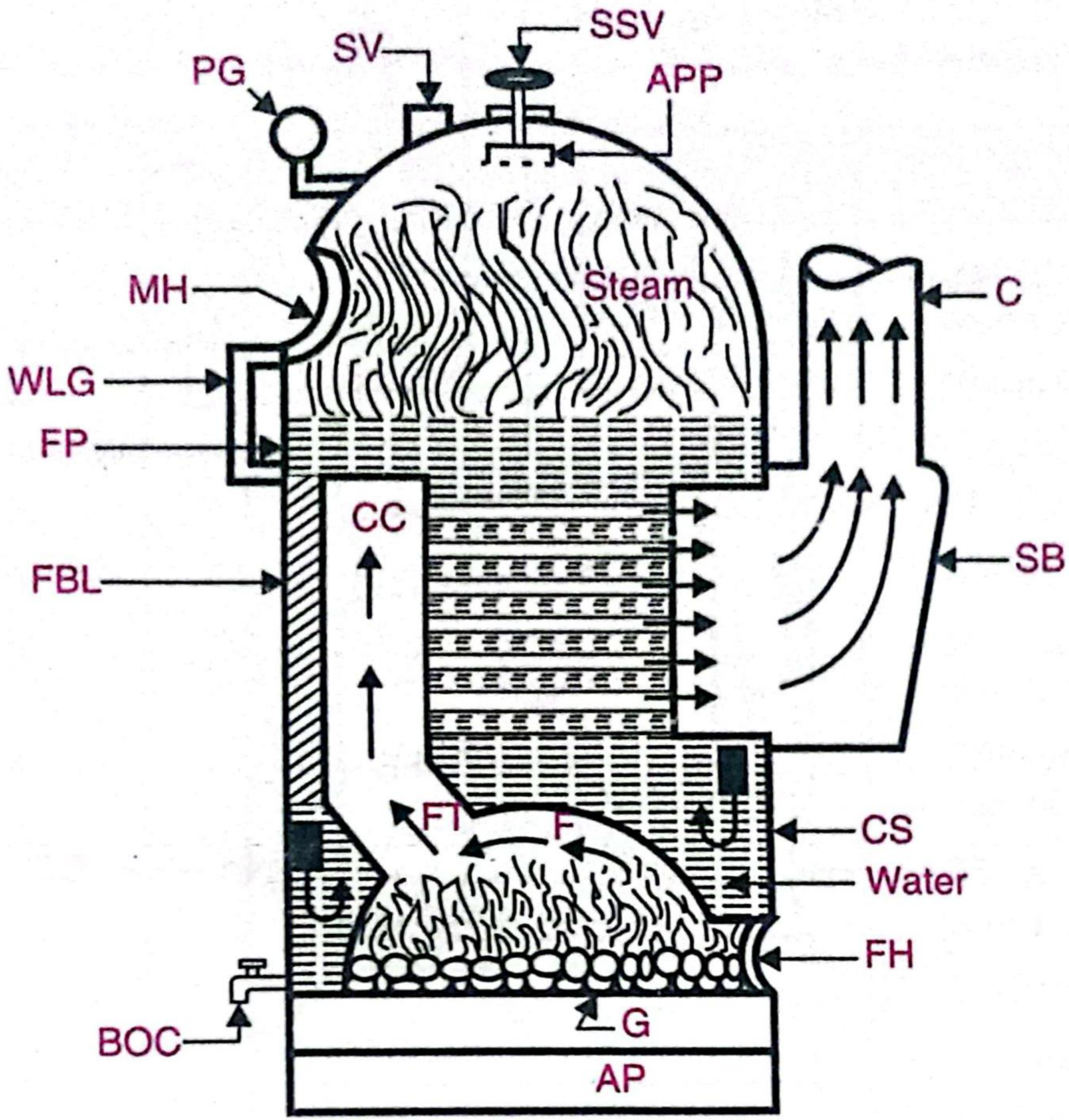


Cylindrical bunker.

Boilers

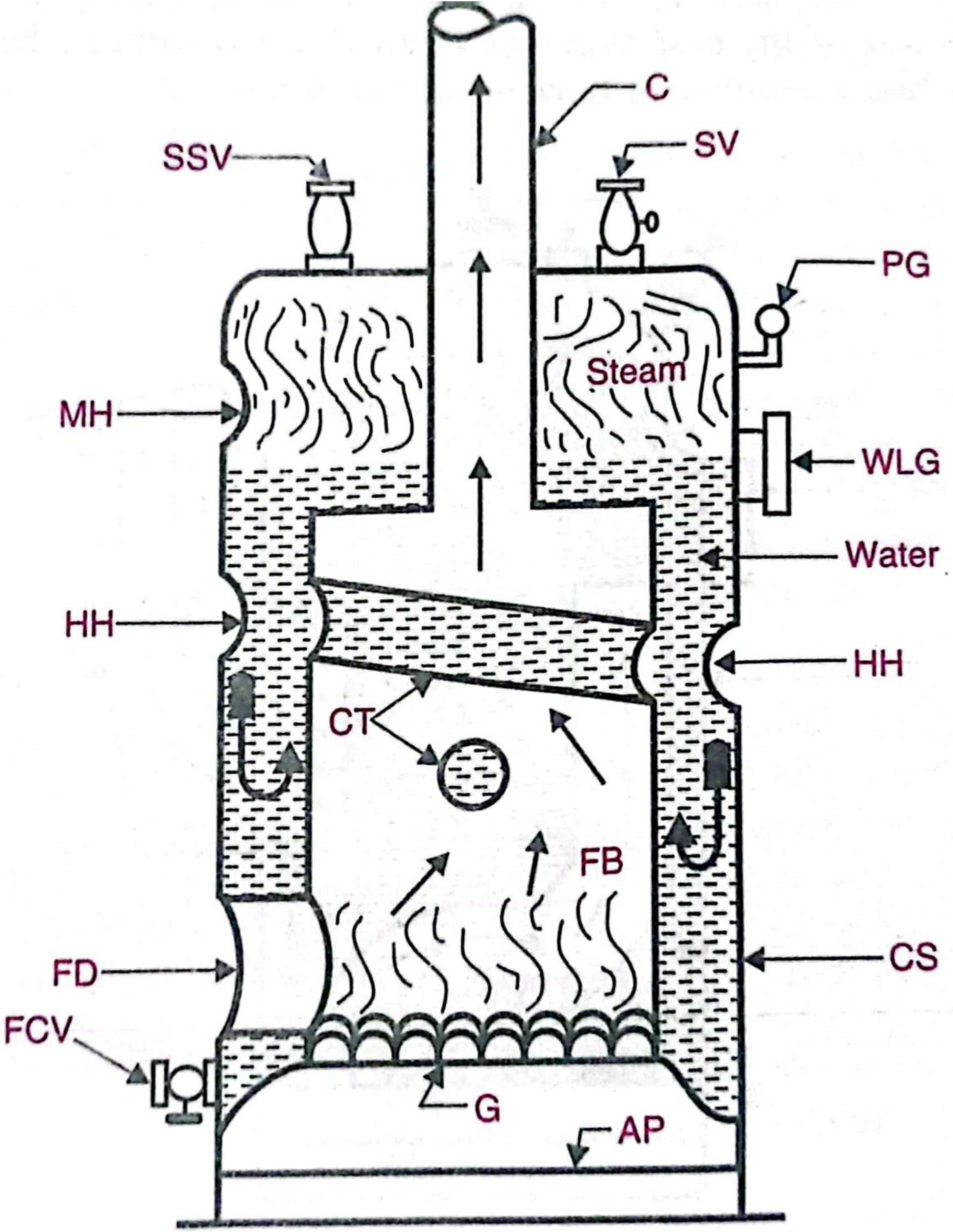
3.14.3. Comparison between 'Fire-tube and Water-tube' Boilers

S. No.	*Particulars*	*Fire-tube boilers*	*Water-tube boilers*
1.	*Position of water and hot gases*	Hot gases inside the tubes and water outside the tubes.	Water inside the tubes and hot gases outside the tubes.
2.	*Mode of firing*	Generally internally fired.	Externally fired.
3.	*Operating pressure*	Operating pressure limited to 16 bar.	Can work under as high pressure as 100 bar.
4.	*Rate of steam production*	Lower	Higher.
5.	*Suitability*	Not suitable for large power plants.	Suitable for large power plants.
6.	*Risk on bursting*	Involves lesser risk on explosion due to lower pressure.	Involves more risk on bursting due to high pressure.
7.	*Floor area*	For a given power it occupies more floor area.	For a given power it occupies less floor-area.
8.	*Construction*	Difficult	Simple
9.	*Transportation*	Difficult	Simple
10.	*Shell diameter*	Large for same power	Small for same power
11.	*Chances of explosion*	Less	More
12.	*Treatment of water*	Not so necessary	More necessary
13.	*Accessibility of various parts*	Various parts not so easily accessible for cleaning, repair and inspection.	Various parts are more accessible.
14.	*Requirement of skill*	Require less skill for efficient and economic working.	Require more skill and careful attention.



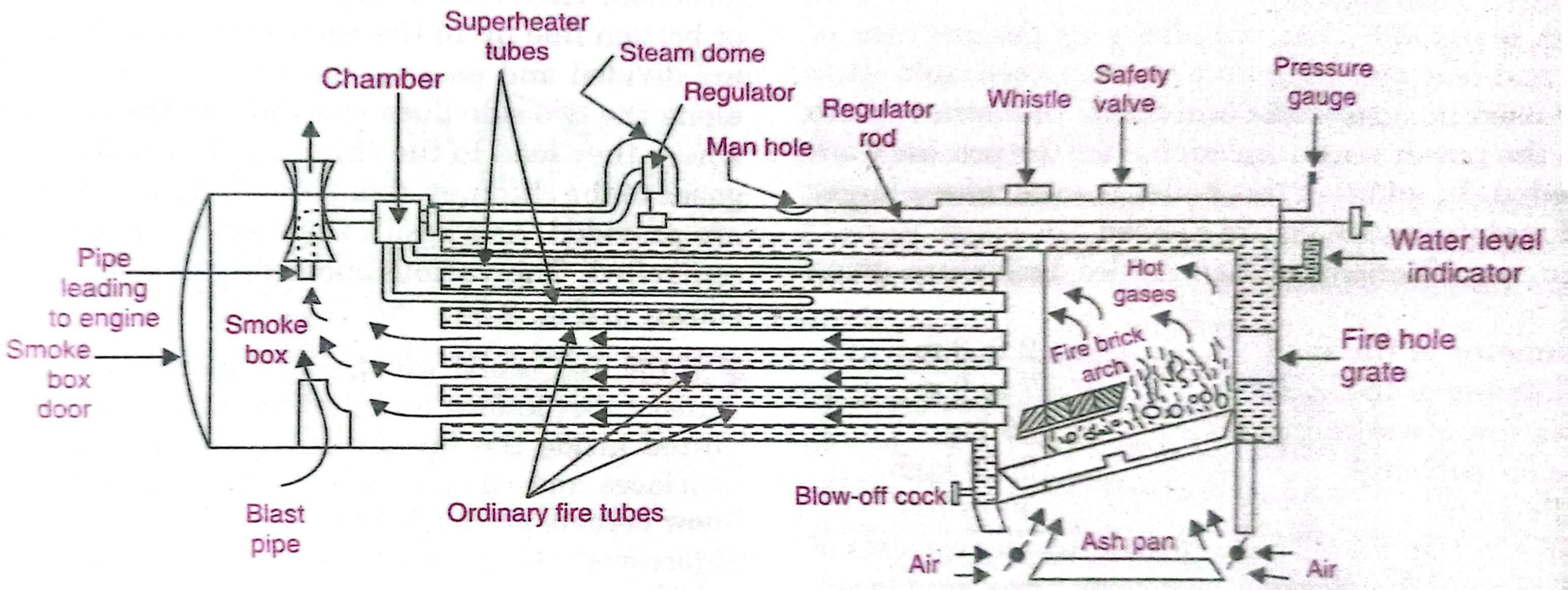
CS = Cylindrical shell	*FT* = Flue tube
CC = Combustion chamber	*SB* = Smoke box
FBL = Fir brick lining	*C* = Chimney
F = Furnace (dome shaped)	*FH* = Fire hole
G = Grate	*BOC* = Blow-off cock
AP = Ash pit	*SSV* = Steam stop valve
SV = Safety valve	*APP* = Antipriming pipe
MH = Man hole	*PG* = Pressure gauge
WLG = Water level gauge	

Cochran Boiler

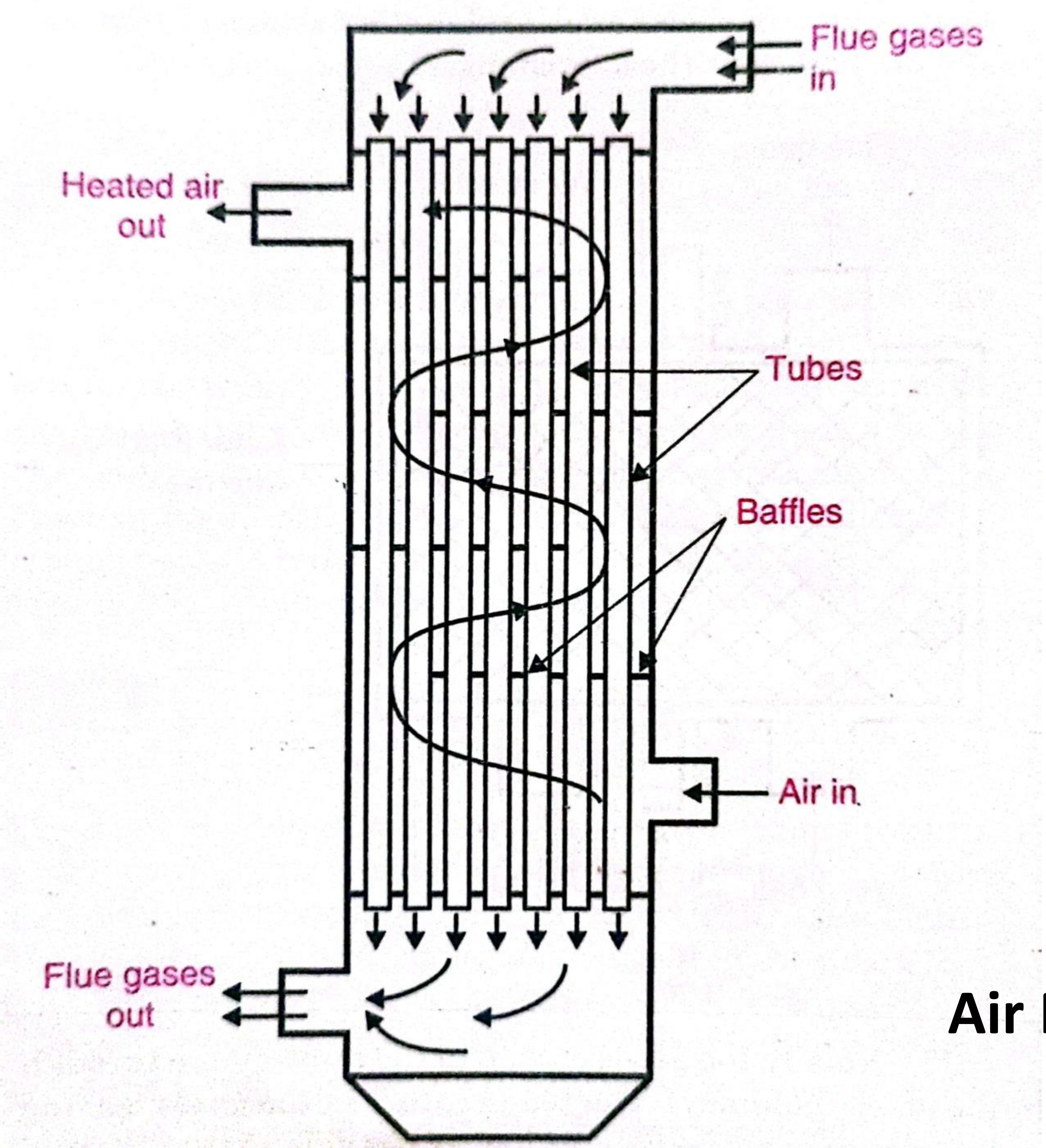


CS = Cylindrical shell	WLG = Water level gauge
MH = Manhole	C = Chimney
CT = Cross tubes	HH = Hand hole
FB = Fire box	FD = Fire door
AP = Ash pit	G = Grate
PG = Pressure gauge	SSV = Steam stop valve
SV = Safety valve	FCV = Feed check valve

Simple vertical boiler



Locomotive boiler



Air Pre-heater

Nuclear Power Plant

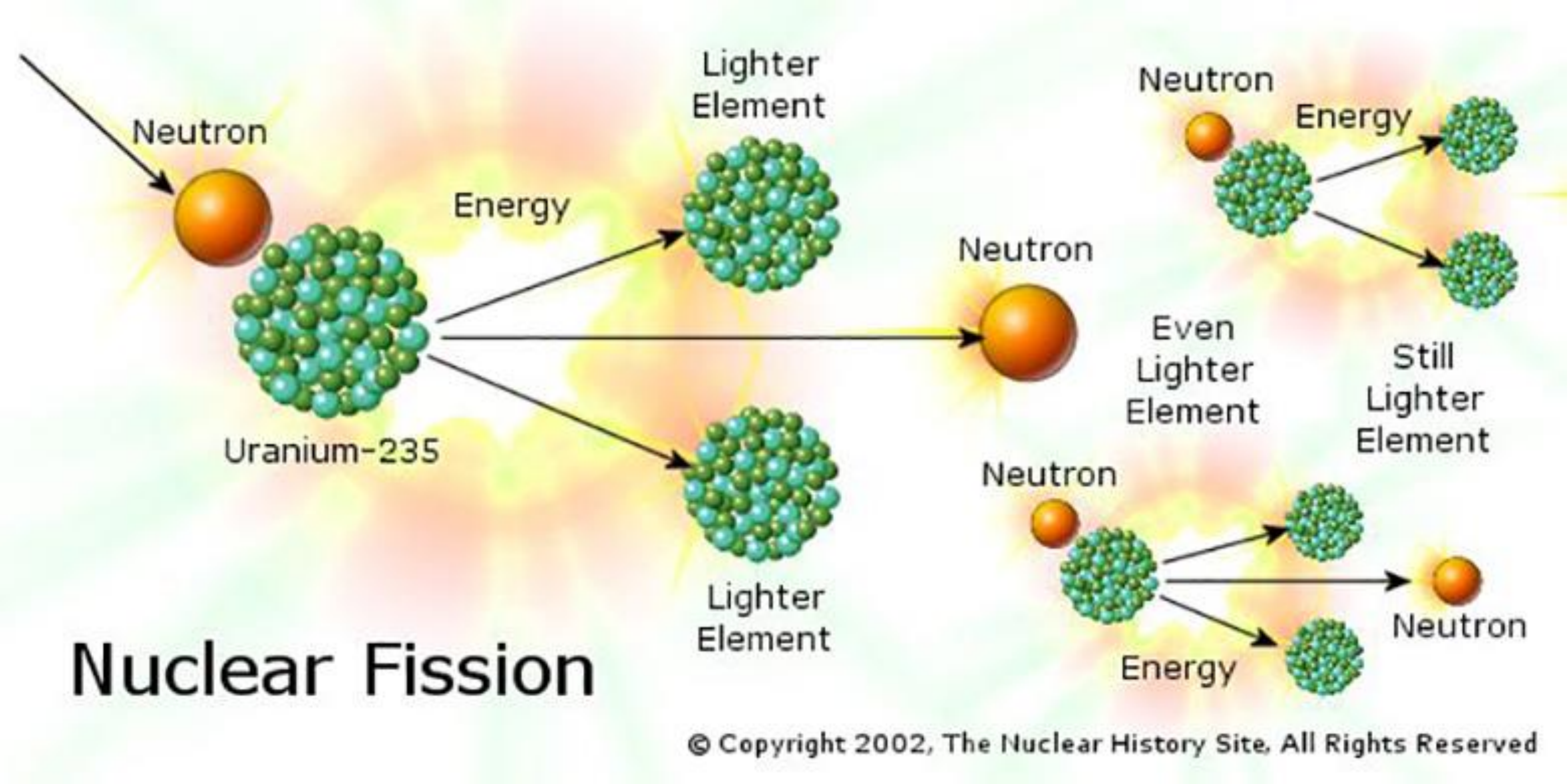
Module-3

Nuclear Fission

- Nuclear fission is a reaction that splits the nucleus of an atom into two or more smaller nuclei.
- This process releases a large amount of energy and often produces gamma photons.
- For example, when a neutron hits a uranium-235 atom, it splits into two smaller nuclei, such as a barium nucleus and a krypton nucleus.
- Nuclear fission can help meet energy needs when chain reactions are controlled in reactors.
- Nuclear power currently provides about 85% of the electricity used.

Case Study: Fission in U-235

- In the nucleus of each atom of uranium-235 (U-235) are 92 protons and 143 neutrons, for a total of 235.
- The arrangement of particles within uranium-235 is somewhat unstable and the nucleus can disintegrate if it is excited by an outside source.
- When a U-235 nucleus absorbs an extra neutron, it quickly breaks into two parts. This process is known as fission.
- Each time a U-235 nucleus splits, it releases two or three neutrons.
- Hence, the possibility exists for creating a chain reaction.


Neutron
Uranium-235
Energy
Lighter Element
Lighter Element
Neutron
Neutron
Energy
Even Lighter Element
Still Lighter Element
Neutron
Neutron
Energy
Nuclear Fission
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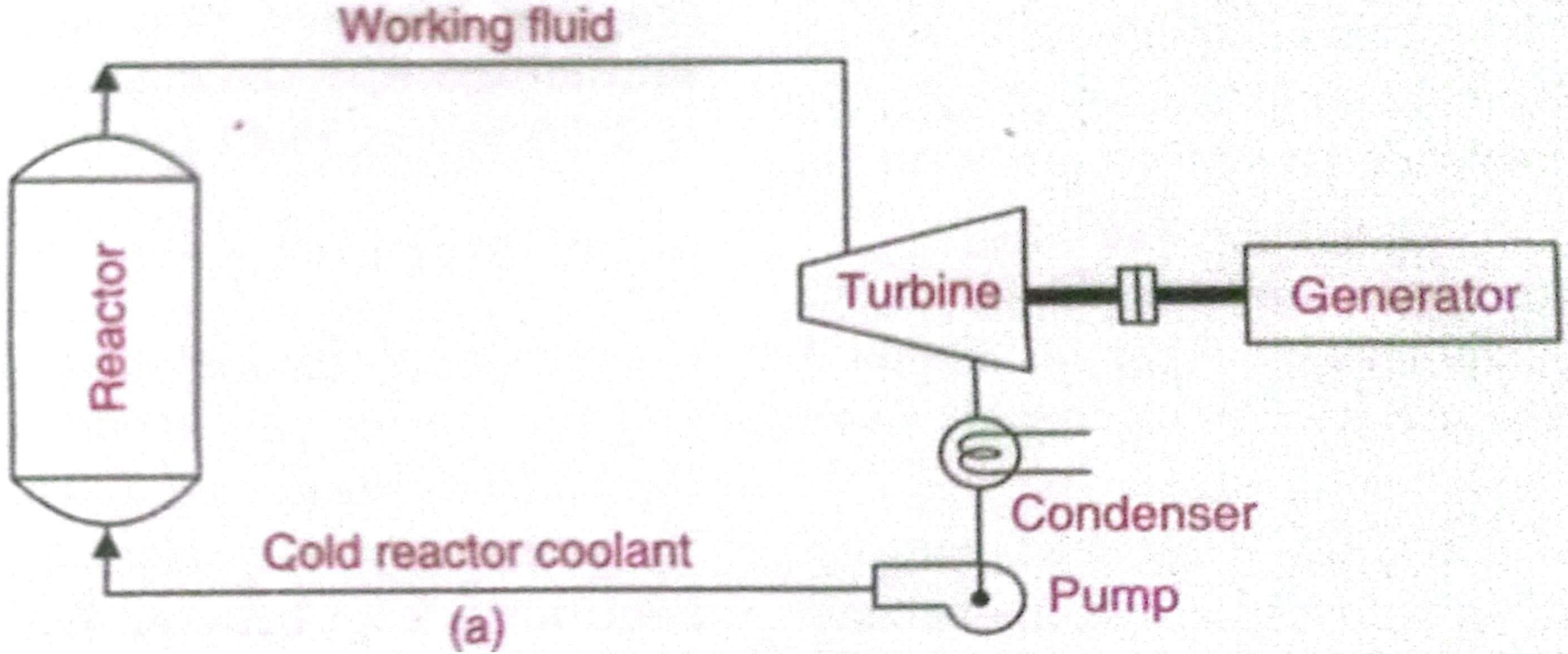
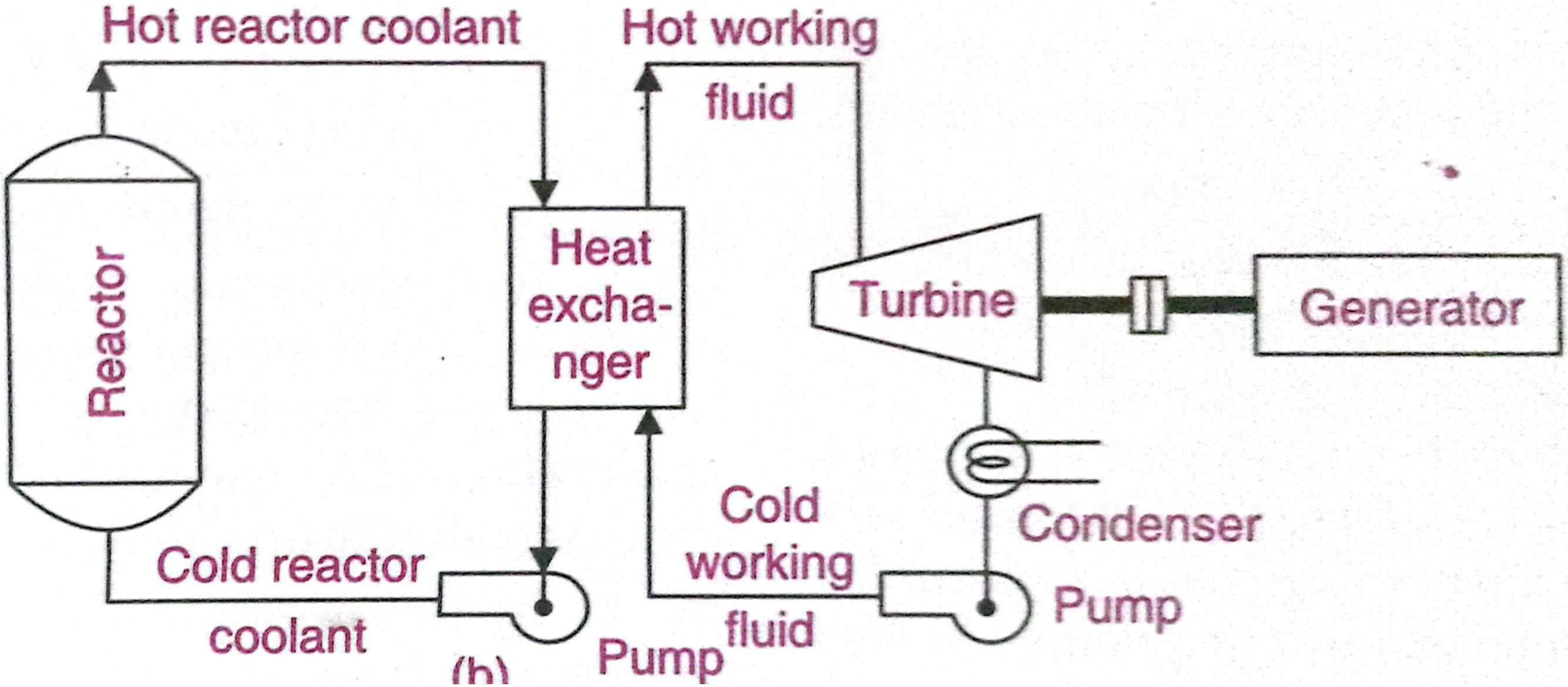
Nuclear Power Systems

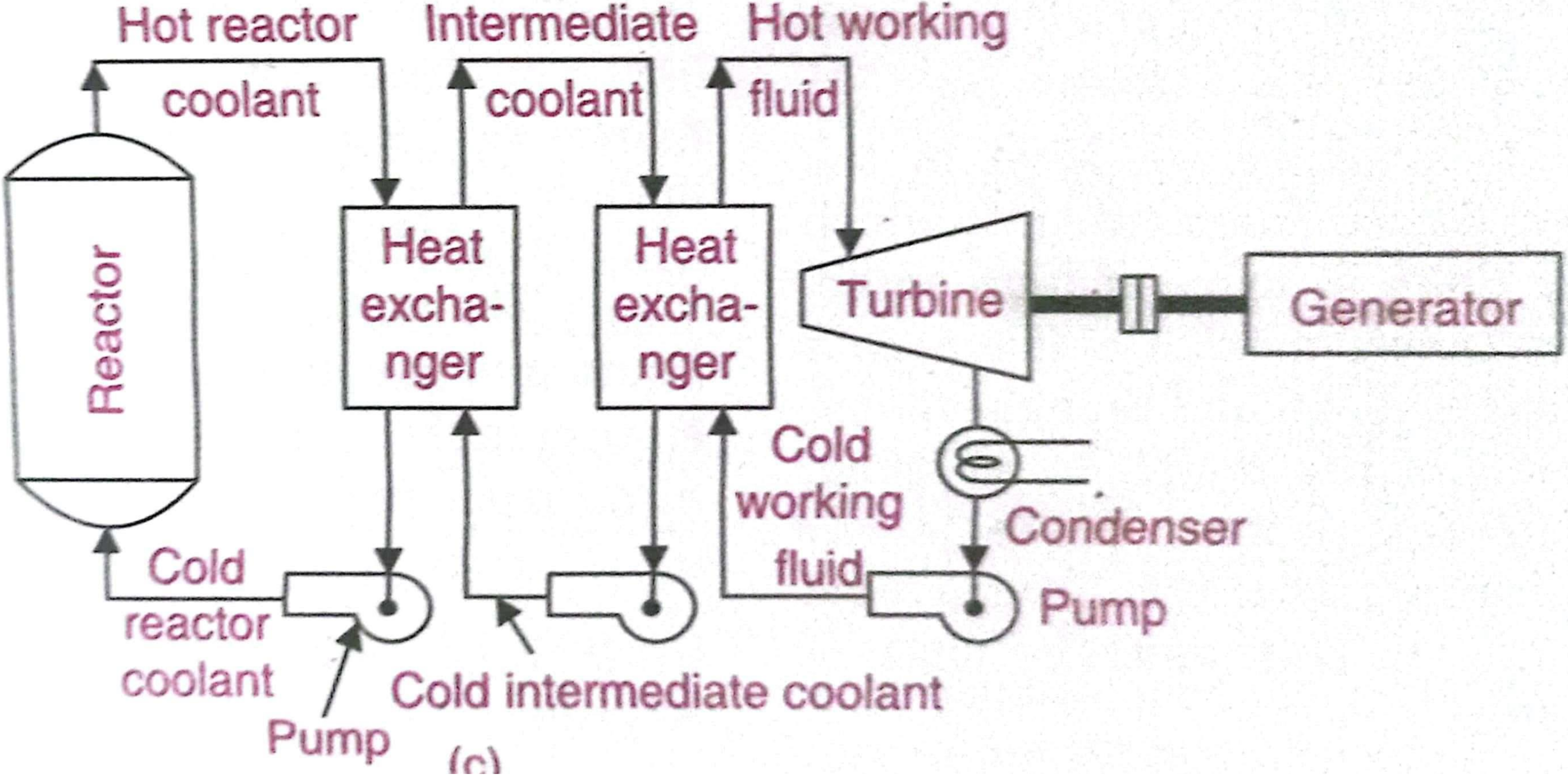
- A nuclear power producing system consist of the following
    1. A controlled fission heat source
    2. A coolant system to remove and transfer the heat produced
    3. Equipment to convert the thermal energy contained in the hot coolant to electric power
- The thermal energy is removed from the heat source by contacting the fuel with a coolant which can be used directly as the working fluid in the power conversion cycle or indirectly to heat another fluid to be used as the working fluid



Direct cycle



Indirect cycle, reactor coolant transfers heat to separate working fluid



Indirect cycle with intermediate loop, reactor coolant transfers heat through intermediate heat transfer loop to working fluid